# FOR A KOSZUL REGULAR IDEAL THE EXT ALGEBRA OF THE QUOTIENT IS THE EXTERIOR ALGEBRA ON ITS FIRST EXT SHEAF

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0BQY (sources/stacks-duality/pinned-original-source.tex line 3661).

*Allowed standard prerequisites (editorial).*

- Koszul regular sheaves of ideals. Divisors, Definition labelled divisors-definition-regular-ideal-sheaf, invoked in the statement. Supplies the hypothesis and the local Koszul regular sequence.
- Composition of internal Homs and the simple tensor product comparison. Cohomology, Lemma labelled cohomology-lemma-internal-hom-composition, cited at proof lines 3678-3680, and More on Algebra, Equation labelled more-algebra-equation-simple-tor-product, cited at proof lines 3688-3692. Fixes the multiplication whose exterior-algebra structure is computed.
- Derived Hom of a strictly perfect complex. Cohomology, Lemma labelled cohomology-lemma-Rhom-strictly-perfect, cited at proof lines 3714-3718. Replaces RSheafHom by the Hom complex of the Koszul dga with vanishing differentials.
- The Koszul complex as a resolution of the quotient. The quasi-isomorphism A^bullet -> O_X built from the divided power polynomial algebra in proof lines 3700-3712. Identifies the Ext sheaves with the Hom sheaves out of the graded pieces A^{-i}.

*Source reference.* Original definition 063D: sources/stacks-chow/tags.txt, source lines 7827–7827. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 0A8V: sources/stacks-chow/tags.txt, source lines 12928–12928. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3685 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original equation 068H: sources/stacks-chow/tags.txt, source lines 8006–8006. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3695 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 08DM: sources/stacks-chow/tags.txt, source lines 10642–10642. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 3726 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $Y$ be a ringed space. Let $\mathcal{I} \subset \mathcal{O}_Y$ be a sheaf of ideals. Set $\mathcal{O}_X = \mathcal{O}_Y/\mathcal{I}$. If $\mathcal{I}$ is Koszul-regular (Divisors, Definition 063D) then composition on $R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$ defines isomorphisms*

$$\wedge^i(\mathcal{E}xt^1_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)) \longrightarrow \mathcal{E}xt^i_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$$

*for all $i$.*

*Proof.* By composition we mean the map

$$R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \otimes^{\mathbf{L}}_{\mathcal{O}_Y} R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \longrightarrow R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$$

of Cohomology, Lemma 0A8V. This induces multiplication maps

$$\mathcal{E}xt^a_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \otimes_{\mathcal{O}_Y} \mathcal{E}xt^b_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \longrightarrow \mathcal{E}xt^{a+b}_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$$

Please compare with More on Algebra, Equation (068H). The statement of the lemma means that the induced map

$$\mathcal{E}xt^1_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \otimes \ldots \otimes \mathcal{E}xt^1_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \longrightarrow \mathcal{E}xt^i_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X)$$

factors through the wedge product and then induces an isomorphism. To see this is true we may work locally on $Y$. Hence we may assume that we have global sections $f_1, \ldots, f_r$ of $\mathcal{O}_Y$ which generate $\mathcal{I}$ and which form a Koszul regular sequence. Denote

$$\mathcal{A} = \mathcal{O}_Y\langle \xi_1, \ldots, \xi_r \rangle$$

the sheaf of strictly commutative differential graded $\mathcal{O}_Y$-algebras which is a (divided power) polynomial algebra on $\xi_1, \ldots, \xi_r$ in degree $-1$ over $\mathcal{O}_Y$ with differential $\mathrm{d}$ given by the rule $\mathrm{d}\xi_i = f_i$. Let us denote $\mathcal{A}^\bullet$ the underlying complex of $\mathcal{O}_Y$-modules which is the Koszul complex mentioned above. Thus the canonical map $\mathcal{A}^\bullet \to \mathcal{O}_X$ is a quasi-isomorphism. We obtain quasi-isomorphisms

$$R\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \to \mathcal{H}om^\bullet(\mathcal{A}^\bullet, \mathcal{A}^\bullet) \to \mathcal{H}om^\bullet(\mathcal{A}^\bullet, \mathcal{O}_X)$$

by Cohomology, Lemma 08DM. The differentials of the latter complex are zero, and hence

$$\mathcal{E}xt^i_{\mathcal{O}_Y}(\mathcal{O}_X, \mathcal{O}_X) \cong \mathcal{H}om_{\mathcal{O}_Y}(\mathcal{A}^{-i}, \mathcal{O}_X)$$

For $j \in \{1, \ldots, r\}$ let $\delta_j : \mathcal{A} \to \mathcal{A}$ be the derivation of degree 1 with $\delta_j(\xi_i) = \delta_{ij}$ (Kronecker delta). A computation shows that $\delta_j \circ \mathrm{d} = -\mathrm{d} \circ \delta_j$ which shows that we get a morphism of complexes

$$\delta_j : \mathcal{A}^\bullet \to \mathcal{A}^\bullet[1].$$

Whence $\delta_j$ defines a section of the corresponding $\mathcal{E}xt$-sheaf. Another computation shows that $\delta_1, \ldots, \delta_r$ map to a basis for $\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{A}^{-1}, \mathcal{O}_X)$ over $\mathcal{O}_X$. Since it is clear that $\delta_j \circ \delta_j = 0$ and $\delta_j \circ \delta_{j'} = -\delta_{j'} \circ \delta_j$ as endomorphisms of $\mathcal{A}$ and hence in the $\mathcal{E}xt$-sheaves we obtain the statement that our map above factors through the exterior power. To see we get the desired isomorphism the reader checks that the elements

$$\delta_{j_1} \circ \ldots \circ \delta_{j_i}$$

for $j_1 < \ldots < j_i$ map to a basis of the sheaf $\mathcal{H}om_{\mathcal{O}_Y}(\mathcal{A}^{-i}, \mathcal{O}_X)$ over $\mathcal{O}_X$. □